VenturePulse

Early-stage startup outcome analytics

Product documentation

Historical startups: 1400 | Best model: Logistic Regression | Balanced accuracy: 0.749 | F1: 0.716

Local web app at http://localhost:8000. All numbers in this document were produced by the local pipeline.

Data note: public download unavailable; results use the realistic synthetic fallback dataset (same schema).

1. VenturePulse overview

VenturePulse helps investors, founders and analysts estimate whether an early-stage startup resembles historically successful or failed companies. Users enter structured signals (funding, rounds, milestones, relationships, industry, state, investor indicators) plus a free-text business description. The system builds a fused representation of structured features and transformer text embeddings, retrieves comparable historical startups with cosine similarity, and predicts with the best validated classifier. Every prediction ships with evidence: similar cases, similarity scores, key features and confidence.

2. Problem being addressed

Early-stage outcomes are uncertain and data is scattered across funding, team, market and narrative signals. VenturePulse unifies these views so a user can compare a new venture against comparable historical cases instead of relying on gut feel alone. Outputs are analytical estimates, not guarantees or investment advice.

3. Product features

- Outcome prediction (SUCCESS / FAILURE) with calibrated probabilities and confidence.
- Similar-case retrieval (k = 10/20/30/50, default 50) with class-balanced evidence and similarity scores.
- Four validated models with a live comparison table, confusion matrix and charts.
- Startup insights: industry success rates, funding distributions, correlations, top features.
- JSON APIs for health, metrics, insights and prediction; clean validation and error handling.
- Local-first: no accounts, no external prediction services, no required API keys.

4. User workflow

- Open the dashboard to review data coverage and the best validated model.
- Go to Predict Startup, fill funding/traction/profile fields and write a business description.
- Press Predict Startup Success and read the estimated outcome, confidence and risk note.
- Inspect key contributing features and the ranked table of similar historical startups.
- Cross-check on Model Performance and Startup Insights before forming a view.

5. Technology stack

Python, FastAPI, Jinja2, HTML5/CSS3, pandas, scikit-learn, XGBoost, sentence-transformers (all-MiniLM-L6-v2, 384-d), torch, joblib, matplotlib, uvicorn, pytest, Playwright. Embeddings backend in this deployment: sentence-transformers.

6. System architecture

Browser UI (Jinja2 templates + static CSS) talks to a FastAPI backend. The backend loads cached artifacts (preprocessor, best model, fused training matrix, training frame, metadata) from models/ at startup; the first start trains them via scripts/train.py. Prediction flows through validation, structured transform, embedding, fusion, cosine retrieval and classification. Charts are precomputed into app/static/img/.

7. Data pipeline

Load public CSV from data/raw/ when available, else generate the realistic synthetic fallback (same schema, seed 42). Map acquired to SUCCESS and closed to FAILURE. Deduplicate, coerce numerics, normalize categoricals, median/most-frequent imputation, log1p funding feature, clean descriptions and mask company names. Split with

StratifiedShuffleSplit (5 x 80/20, seed 42); preprocessing is fit on train only. Dataset: 1400 records (595 SUCCESS / 805 FAILURE).

8. Machine-learning pipeline

- LogisticRegression (lbfgs, balanced, C=1.0, max_iter=1000).
- DecisionTreeClassifier (max_depth=8, min_samples_split=10, min_samples_leaf=5, balanced).
- RandomForestClassifier (300 trees, balanced, min_samples_leaf=2).
- XGBClassifier (400 trees, depth 6, lr 0.05, subsample/colsample 0.8, reg_lambda 1.0).
- Primary metric: balanced accuracy; also accuracy, precision, recall, F1, confusion matrix (mean +/- std).

9. Similarity retrieval mechanism

Structured features are scaled (StandardScaler + one-hot) then L2-normalized; text embeddings are L2-normalized; the fused vector is concat(0.5 * structured, 0.5 * text) with a final L2 normalization (alpha = 0.5). Cosine similarity is computed against training records only, sorted descending, and filled to approximately half SUCCESS and half FAILURE (graceful fallback when a class is short). Scores are shown unmodified; the target label is never used during retrieval.

10. Model comparison

Logistic Regression [SELECTED]: balanced_acc=0.749 +/- 0.025, acc=0.751, prec=0.696, rec=0.739, f1=0.716.

Decision Tree: balanced_acc=0.614 +/- 0.041, acc=0.621, prec=0.552, rec=0.568, f1=0.559.

Random Forest: balanced_acc=0.655 +/- 0.021, acc=0.662, prec=0.603, rec=0.607, f1=0.604.

XGBoost: balanced_acc=0.694 +/- 0.011, acc=0.710, prec=0.684, rec=0.590, f1=0.634.

Representation study (balanced accuracy): structured_only=0.714 +/- 0.025, text_only=0.599 +/- 0.013, combined=0.749 +/- 0.025. Combining structured and textual information performs best. Similar-case weighted vote: k=10=0.654, k=20=0.682, k=30=0.691, k=50=0.698 (default k=50).

11. Evaluation methodology

StratifiedShuffleSplit, 5 repetitions, 80% train / 20% test, seed 42. Metrics computed from actual predictions on held-out splits; reported as mean +/- standard deviation. No target leakage: outcome columns are excluded from features, preprocessing is fit on train only, and retrieval never sees test records.

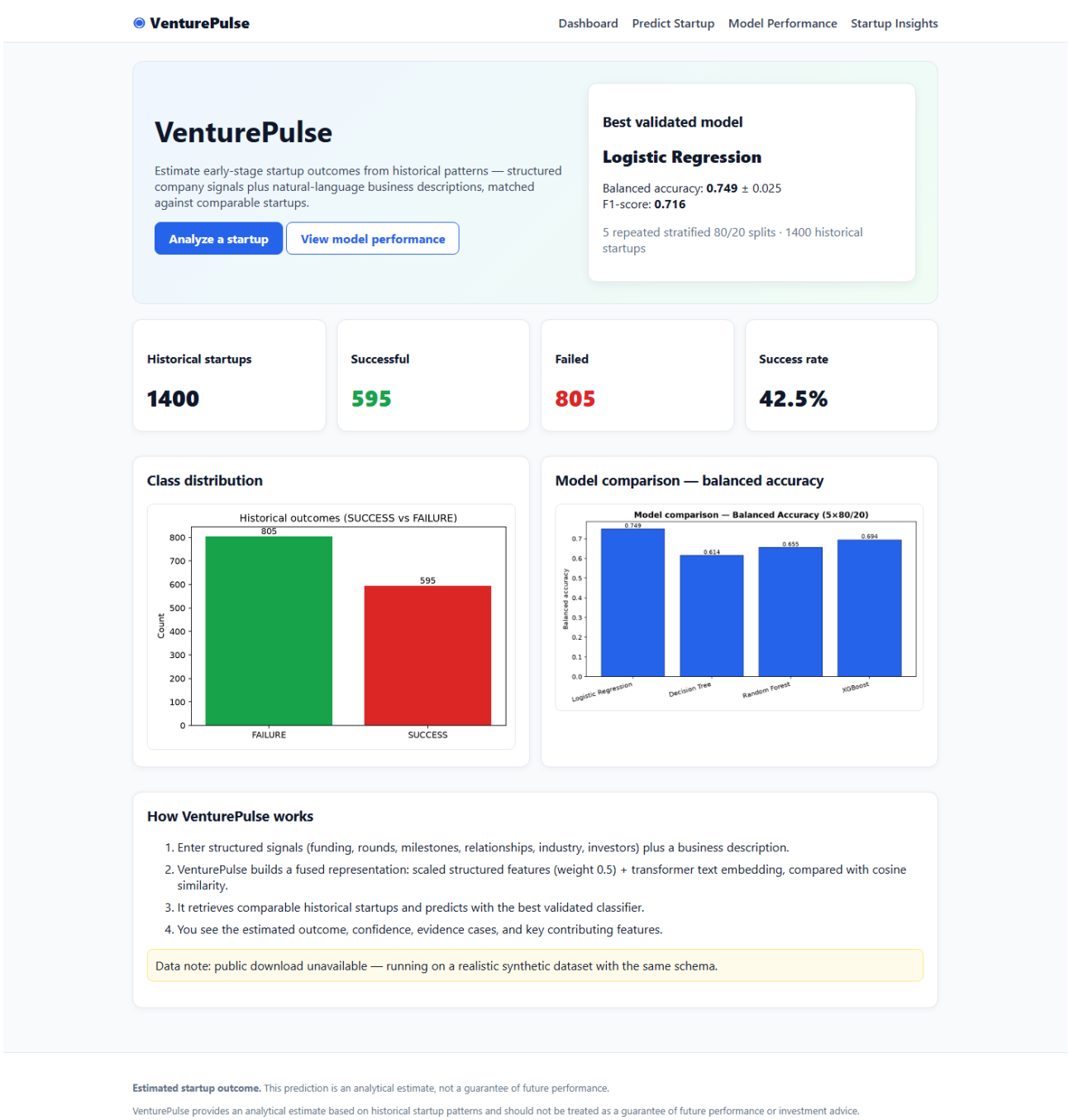
12. Actual results and metrics

Best model: Logistic Regression with balanced accuracy 0.749 +/- 0.025 and F1 0.716. Held-out confusion matrix (last split, rows actual FAILURE/SUCCESS, cols predicted): [[124, 37], [29, 90]].

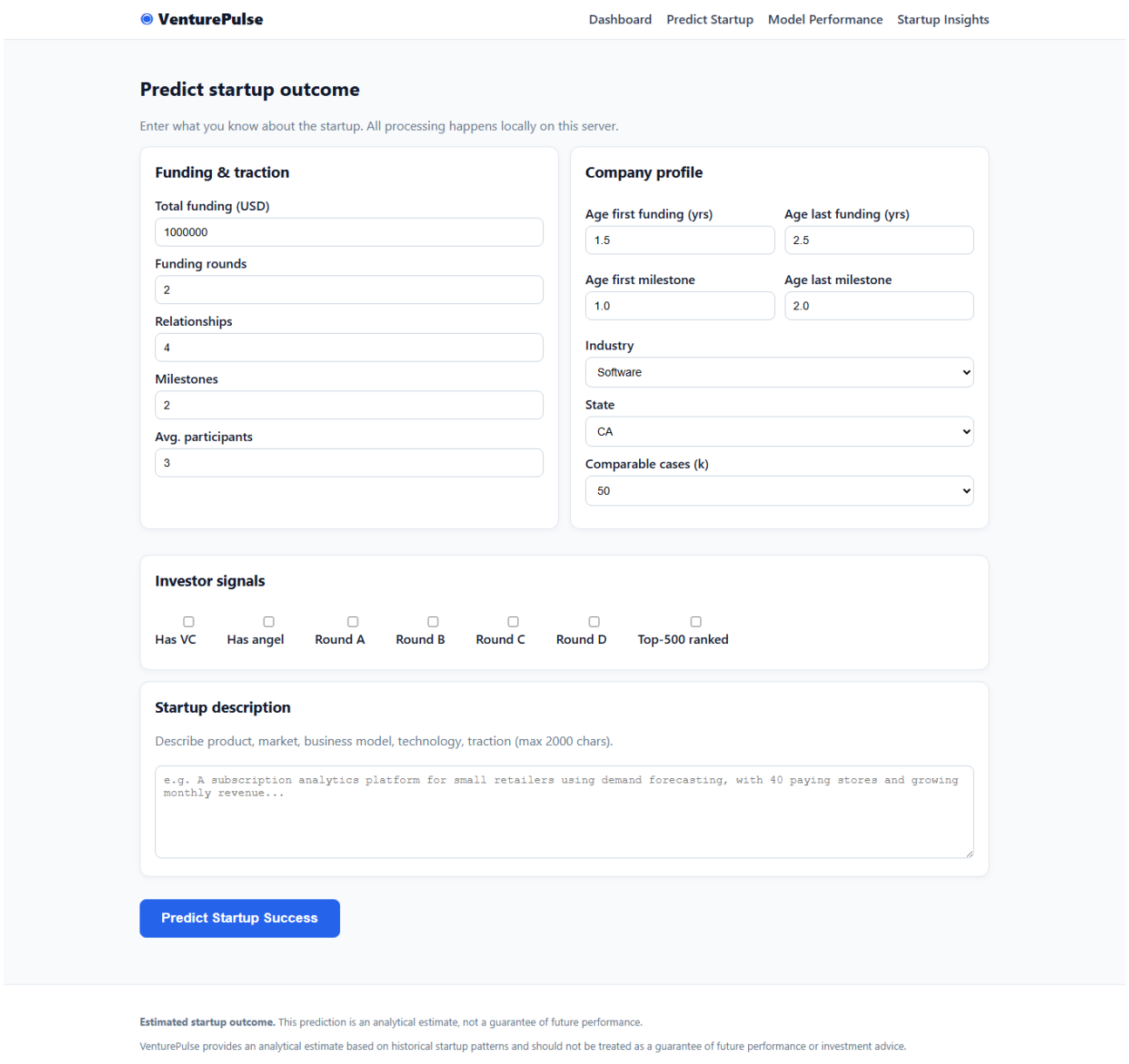
13. Screenshots

Captured from the running application at http://localhost:8000.

01-dashboard.png



02-predict-form.png



03-prediction-result.png

VenturePulse Dashboard Predict Startup Model Performance Startup Insights

Estimated startup outcome

SUCCESS Confidence 65.9% · Model: Logistic Regression · Moderate risk

Success probability **65.9%** · Failure probability **34.1%**

This prediction is an analytical estimate, not a guarantee of future performance.

Key contributing features

- has_roundC: value 0.00 (scaled), weight 2.495 — supports FAILURE
- funding_rounds: value -0.07 (scaled), weight 2.401 — supports FAILURE
- relationships: value -0.22 (scaled), weight 2.380 — supports FAILURE
- has_VC: value 1.00 (scaled), weight 2.348 — supports SUCCESS
- milestones: value -0.09 (scaled), weight 2.243 — supports FAILURE
- Retrieved 50 comparable historical startups (25 SUCCESS / 25 FAILURE); top similarity 0.570.
- Average similarity to SUCCESS cases: 0.475; to FAILURE cases: 0.484.

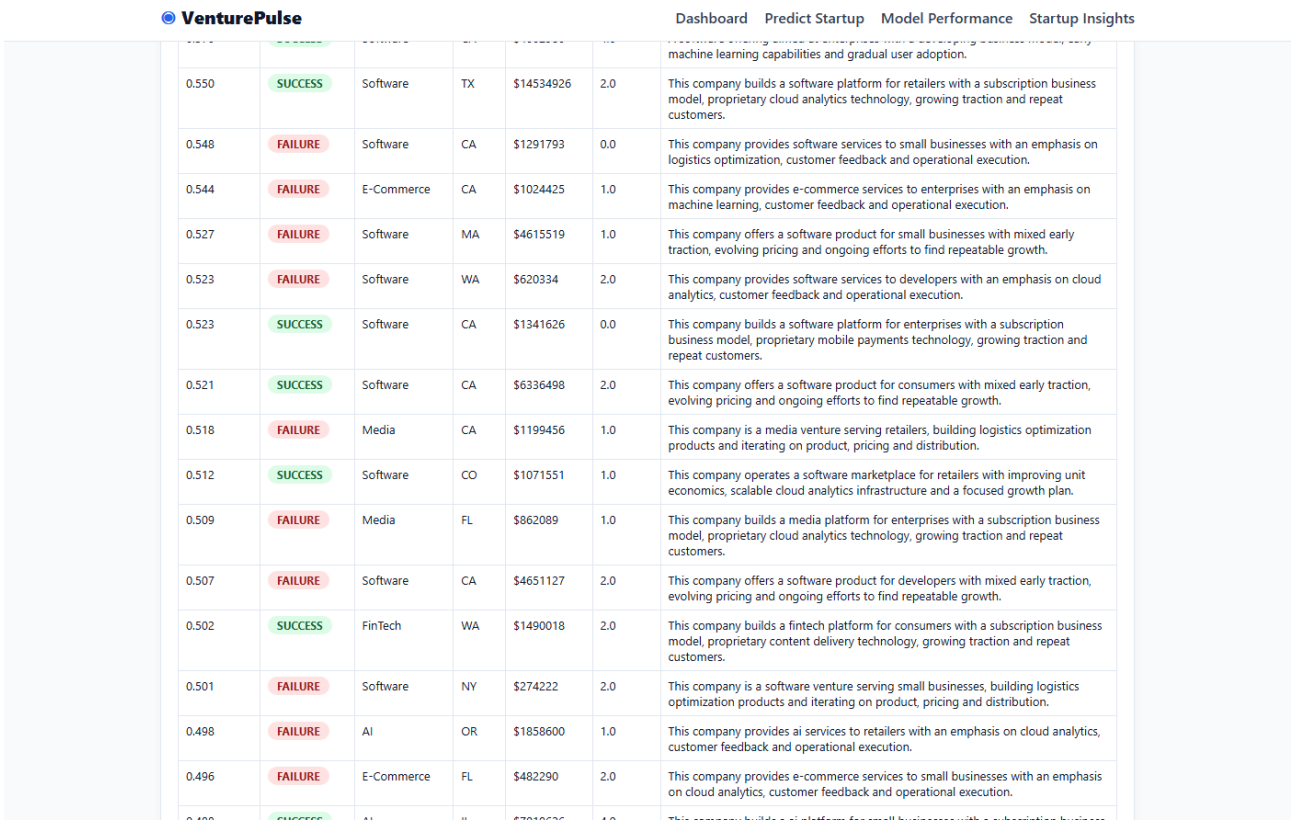
Your inputs

- Industry: Software · State: CA
- Funding: $2500000.0 across 2.0 rounds
- Relationships 4.0 · Milestones 2.0
- A subscription analytics platform for small retailers using demand forecasting, with 40 paying stores, growing monthly revenue and partnerships with two distributors.

Similar historical startups (k=50 · 25 SUCCESS / 25 FAILURE)

Similarity	Outcome	Industry	State	Funding	Rounds	Summary
0.570	SUCCESS	Software	CA	$4062980	4.0	A software offering aimed at enterprises with a developing business model, early machine learning capabilities and gradual user adoption.
0.550	SUCCESS	Software	TX	$14534926	2.0	This company builds a software platform for retailers with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.548	FAILURE	Software	CA	$1291793	0.0	This company provides software services to small businesses with an emphasis on logistics optimization, customer feedback and operational execution.
0.544	FAILURE	E-Commerce	CA	$1024425	1.0	This company provides e-commerce services to enterprises with an emphasis on machine learning, customer feedback and operational execution.
0.527	FAILURE	Software	MA	$4615519	1.0	This company offers a software product for small businesses with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.523	FAILURE	Software	WA	$620334	2.0	This company provides software services to developers with an emphasis on cloud analytics, customer feedback and operational execution.
0.523	SUCCESS	Software	CA	$1341626	0.0	This company builds a software platform for enterprises with a subscription business model, proprietary mobile payments technology, growing traction and repeat customers.
0.521	SUCCESS	Software	CA	$6336498	2.0	This company offers a software product for consumers with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.518	FAILURE	Media	CA	$1199456	1.0	This company is a media venture serving retailers, building logistics optimization products and iterating on product, pricing and distribution.
0.512	SUCCESS	Software	CO	$1071551	1.0	This company operates a software marketplace for retailers with improving unit economics, scalable cloud analytics infrastructure and a focused growth plan.
0.509	FAILURE	Media	FL	$862089	1.0	This company builds a media platform for enterprises with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.507	FAILURE	Software	CA	$4651127	2.0	This company offers a software product for developers with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.502	SUCCESS	FinTech	WA	$1490018	2.0	This company builds a fintech platform for consumers with a subscription business model, proprietary content delivery technology, growing traction and repeat customers.
0.501	FAILURE	Software	NY	$274222	2.0	This company is a software venture serving small businesses, building logistics optimization products and iterating on product, pricing and distribution.
0.498	FAILURE	AI	OR	$1858600	1.0	This company provides ai services to retailers with an emphasis on cloud analytics, customer feedback and operational execution.
0.496	FAILURE	E-Commerce	FL	$482290	2.0	This company provides e-commerce services to small businesses with an emphasis on cloud analytics, customer feedback and operational execution.
0.488	SUCCESS	AI	IL	$7019626	4.0	This company builds a ai platform for small businesses with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.482	FAILURE	CleanTech	CA	$1038077	2.0	This company is a cleantech venture serving retailers, building cloud analytics products and iterating on product, pricing and distribution.
0.479	FAILURE	E-Commerce	CA	$815338	2.0	This company is a e-commerce venture serving consumers, building machine learning products and iterating on product, pricing and distribution.
0.475	SUCCESS	Software	MA	$2659596	2.0	This company builds a software platform for small businesses with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.475	SUCCESS	Software	FL	$2046875	4.0	A software startup focused on retailers, developing logistics optimization solutions and pursuing partnerships and customer growth.
0.475	FAILURE	Software	TX	$1129824	0.0	A software offering aimed at consumers with a developing business model, early machine learning capabilities and gradual user adoption.
0.473	FAILURE	Media	CA	$3575683	3.0	This company built a media tool for retailers and is working through positioning, pricing and partnership challenges.
0.471	SUCCESS	Mobile	PA	$2655045	2.0	This company builds a mobile platform for small businesses with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.470	FAILURE	Software	FL	$3639531	1.0	This company builds a software platform for schools with a subscription business model, proprietary machine learning technology, growing traction and repeat customers.
0.469	SUCCESS	Software	TX	$35714927	4.0	This company offers a software product for retailers with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.469	SUCCESS	FinTech	WA	$5009665	2.0	This company operates a fintech marketplace for schools with improving unit economics, scalable cloud analytics infrastructure and a focused growth plan.

06-similar-startups.png



						machine learning capabilities and gradual user adoption.
0.550	SUCCESS	Software	TX	$14534926	2.0	This company builds a software platform for retailers with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.548	FAILURE	Software	CA	$1291793	0.0	This company provides software services to small businesses with an emphasis on logistics optimization, customer feedback and operational execution.
0.544	FAILURE	E-Commerce	CA	$1024425	1.0	This company provides e-commerce services to enterprises with an emphasis on machine learning, customer feedback and operational execution.
0.527	FAILURE	Software	MA	$4615519	1.0	This company offers a software product for small businesses with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.523	FAILURE	Software	WA	$620334	2.0	This company provides software services to developers with an emphasis on cloud analytics, customer feedback and operational execution.
0.523	SUCCESS	Software	CA	$1341626	0.0	This company builds a software platform for enterprises with a subscription business model, proprietary mobile payments technology, growing traction and repeat customers.
0.521	SUCCESS	Software	CA	$6336498	2.0	This company offers a software product for consumers with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.518	FAILURE	Media	CA	$1199456	1.0	This company is a media venture serving retailers, building logistics optimization products and iterating on product, pricing and distribution.
0.512	SUCCESS	Software	CO	$1071551	1.0	This company operates a software marketplace for retailers with improving unit economics, scalable cloud analytics infrastructure and a focused growth plan.
0.509	FAILURE	Media	FL	$862089	1.0	This company builds a media platform for enterprises with a subscription business model, proprietary cloud analytics technology, growing traction and repeat customers.
0.507	FAILURE	Software	CA	$4651127	2.0	This company offers a software product for developers with mixed early traction, evolving pricing and ongoing efforts to find repeatable growth.
0.502	SUCCESS	FinTech	WA	$1490018	2.0	This company builds a fintech platform for consumers with a subscription business model, proprietary content delivery technology, growing traction and repeat customers.
0.501	FAILURE	Software	NY	$274222	2.0	This company is a software venture serving small businesses, building logistics optimization products and iterating on product, pricing and distribution.
0.498	FAILURE	AI	OR	$1858600	1.0	This company provides ai services to retailers with an emphasis on cloud analytics, customer feedback and operational execution.
0.496	FAILURE	E-Commerce	FL	$482290	2.0	This company provides e-commerce services to small businesses with an emphasis on cloud analytics, customer feedback and operational execution.

04-model-performance.png

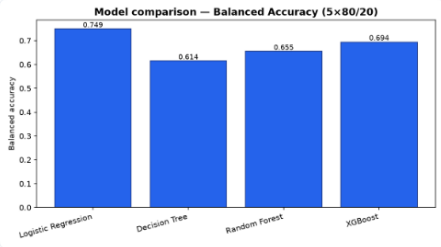
VenturePulse Dashboard Predict Startup Model Performance Startup Insights

Model performance

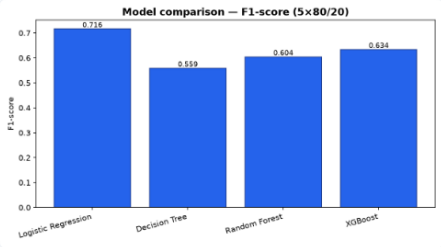
5 repeated stratified 80/20 splits · balanced accuracy is the primary metric · best model: **Logistic Regression**

Model	Balanced acc.	Accuracy	Precision	Recall	F1
Logistic Regression	0.749 ± 0.025	0.751 ± 0.027	0.696 ± 0.038	0.739 ± 0.015	0.716 ± 0.025
Decision Tree	0.614 ± 0.041	0.621 ± 0.037	0.552 ± 0.038	0.568 ± 0.079	0.559 ± 0.056
Random Forest	0.655 ± 0.021	0.662 ± 0.022	0.603 ± 0.031	0.607 ± 0.033	0.604 ± 0.023
XGBoost	0.694 ± 0.011	0.710 ± 0.012	0.684 ± 0.018	0.590 ± 0.013	0.634 ± 0.013

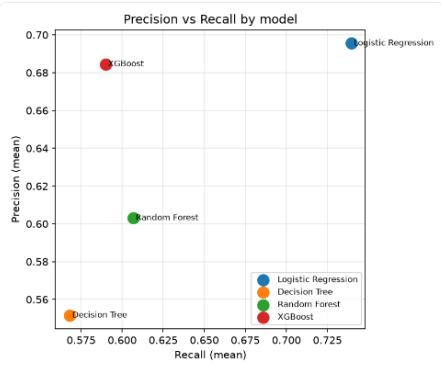
Balanced accuracy



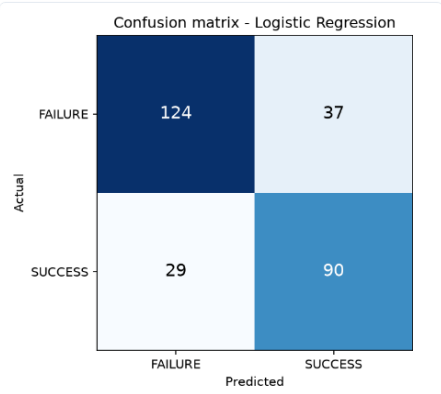
F1-score



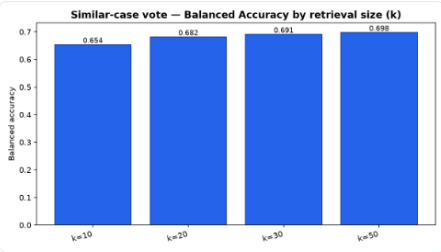
Precision vs recall



Confusion matrix — Logistic Regression

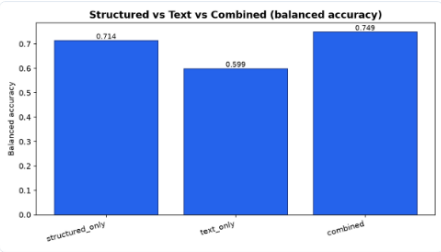


Retrieval size comparison



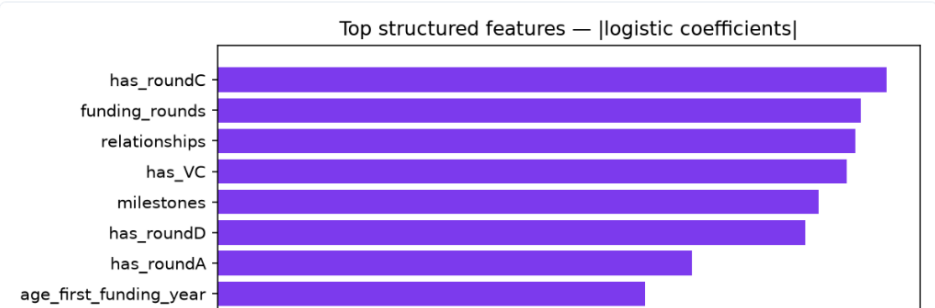
k	Balanced acc.
k=10	0.654 ± 0.009
k=20	0.682 ± 0.013
k=30	0.691 ± 0.016
k=50	0.698 ± 0.012

Structured vs text vs combined

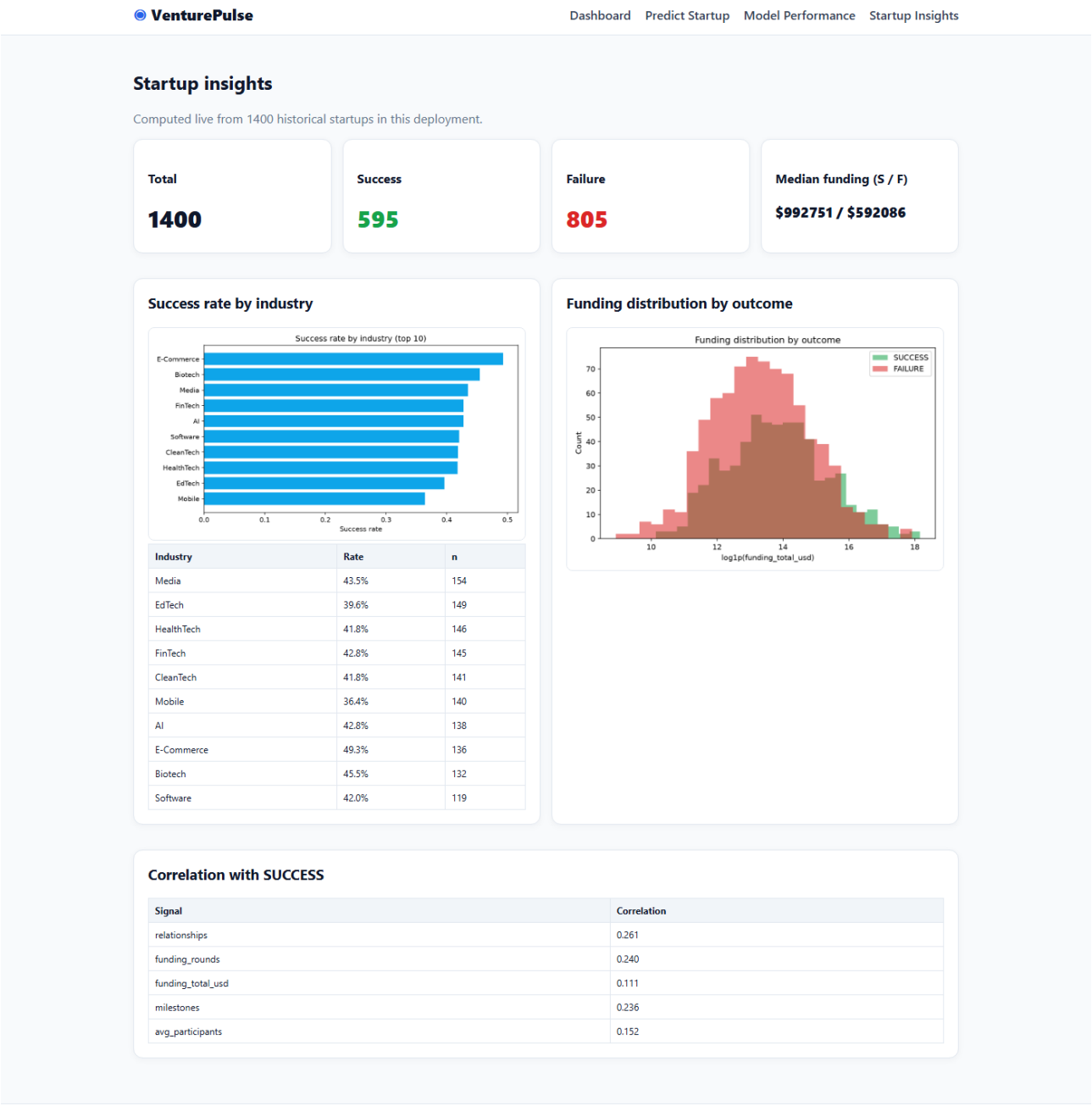


Representation	Balanced acc.
structured_only	0.714 ± 0.025
text_only	0.599 ± 0.013
combined	0.749 ± 0.025

Feature importance



05-insights.png



Industry	Rate	n
Media	43.5%	154
EdTech	39.6%	149
HealthTech	41.8%	146
FinTech	42.8%	145
CleanTech	41.8%	141
Mobile	36.4%	140
AI	42.8%	138
E-Commerce	49.3%	136
Biotech	45.5%	132
Software	42.0%	119

Signal	Correlation
relationships	0.261
funding_rounds	0.240
funding_total_usd	0.111
milestones	0.236
avg_participants	0.152

14. API overview

- GET /api/health -> {status, app, records, best_model}.
- GET /api/model-metrics -> full evaluation results JSON.
- GET /api/insights -> dataset insights JSON.
- POST /predict (JSON) -> {outcome, probability_success, confidence, model, similar[], explanation[]}.
- Pages: GET /, /dashboard, /predict, /performance, /insights; form posts to POST /predict-form.

15. Installation and execution

See HOW_TO_RUN.md. Fastest path: double-click run.bat, wait for dependency install and first-run training, use

http://localhost:8000, stop with Ctrl+C. Docker alternative: run-docker.bat (Recommended if Docker Desktop is installed) or docker build/run commands. Manual: create venv, pip install -r requirements.txt, python -m uvicorn app.main:app --host 127.0.0.1 --port 8000.

16. Limitations

- Estimates reflect historical correlations, not causal guarantees; not investment advice.
- Description input capped at 2000 characters; CPU embeddings.
- Synthetic fallback used when the public CSV is unavailable (recorded in metadata).
- Rare industries/states have small samples; treat their rates cautiously.

17. Practical interpretation

Use the predicted outcome together with confidence and the similar-case table. High confidence with similar cases pointing the same way is more convincing than a bare label. Disagreement between the model and the retrieved evidence is itself informative: inspect the key-feature notes and the funding/milestone profile before deciding what to investigate next.